

Graphing Quadratics – 3-Day Notes and Practice

4.5 Converting from Standard Form to Vertex Form

Goal: Rewrite $y = ax^2 + bx + c$ in vertex form $y = a(x - h)^2 + k$ by completing the square.

Steps

1. Identify a, b, c from the standard form. 2. Factor a out of the quadratic and linear terms. 3. Add and subtract $\left(\frac{b}{2a}\right)^2$ inside the parentheses. 4. Simplify the constants and rewrite as $y = a(x - h)^2 + k$.

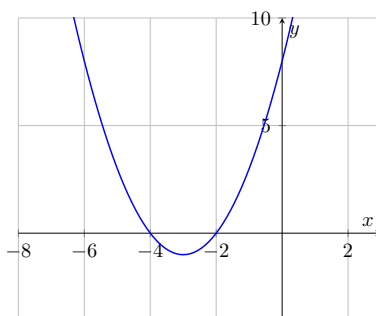
Example 1

$$y = x^2 + 6x + 8$$

$$y = (x^2 + 6x + \underline{\hspace{2cm}}) - \underline{\hspace{2cm}} + 8$$

$$y = (x + 3)^2 - 1$$

Vertex: $(-3, -1)$ **Axis of symmetry:** $x = -3$



Practice A — Convert Each to Vertex Form

1. $y = x^2 + 4x + 1$

Vertex: _____

2. $y = x^2 - 2x - 5$

Vertex: _____

3. $y = 2x^2 + 8x + 3$

Vertex: _____

Challenge: What happens if a is negative? Try $y = -x^2 - 4x - 1$.

4.6 Graphing from Vertex Form

Goal: Use $y = a(x - h)^2 + k$ to graph quickly.

Key Features

Equation: $y = a(x - h)^2 + k$

Vertex: (h, k)

Opens up if $a > 0$, down if $a < 0$

k moves graph up/down

Attributes:

Axis of symmetry: $x = h$

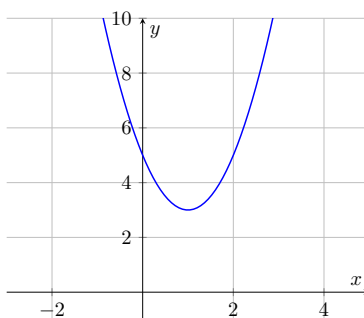
Wider/narrower: $|a| < 1 \rightarrow$ wider, $|a| > 1 \rightarrow$ narrower

h moves graph left/right

Example 2

$$y = 2(x - 1)^2 + 3$$

Vertex: $(1, 3)$ **A.O.S.:** $x = 1$ **Opens:** Up (narrower)



Practice B — Identify Features and Sketch

1. $y = (x + 2)^2 - 1$
2. $y = -3(x - 4)^2 + 2$
3. $y = \frac{1}{2}(x - 1)^2 - 5$

For each: Find vertex, axis of symmetry, direction, and stretch/compression, then sketch.

4.5a Transformations of Quadratics

Goal: Describe how parameters a, h, k transform the graph of $y = x^2$.

Transformation Summary

$$y = a(x - h)^2 + k$$

Change	Effect on Graph
h increases	Shifts right
h decreases	Shifts left
k increases	Shifts up
k decreases	Shifts down
$a < 0$	Reflects across x-axis
$ a > 1$	Narrower (vertical stretch)
$0 < a < 1$	Wider (vertical compression)

Example 3

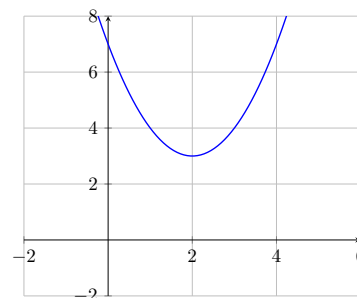
Compare each to $y = x^2$:

Equation and Description

Graph

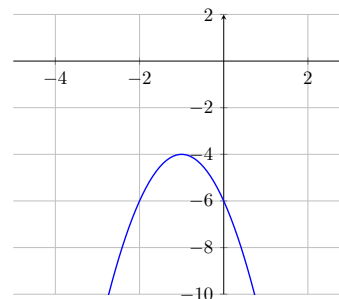
$$y = (x - 2)^2 + 3$$

Right 2, up 3. Vertex (2, 3).



$$y = -2(x + 1)^2 - 4$$

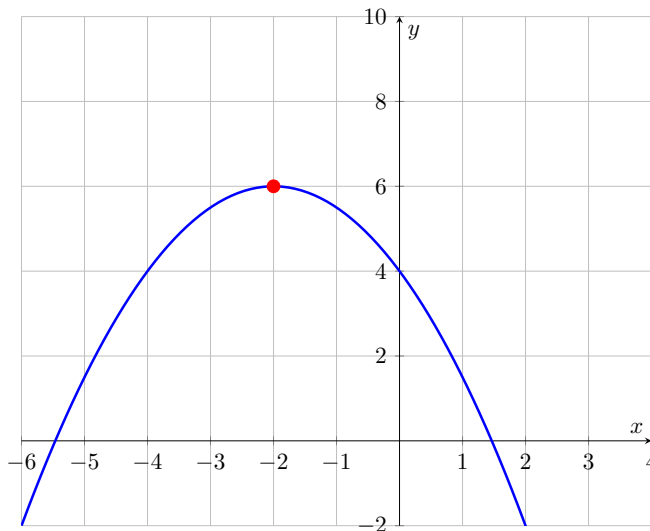
Reflect, narrower, left 1, down 4.



Practice C — Describe Each Transformation and Sketch

1. $y = (x - 5)^2 - 1$
2. $y = -\frac{1}{2}(x + 3)^2 + 2$
3. $y = 4(x - 2)^2$
4. $y = (x + 1)^2 + 6$

Challenge: Given a graph, can you write the equation in vertex form?



Equation in vertex form: _____